DEPARTMENT OF COMPUTER SCIENCE 2023-2024

COLLEGE OF APPLIED SCIENCE

PAYYANUR,NERUVAMBRAM

(Affiliated to Kannur University)

MANAGED BY

INSTITUTE OF HUMAN RESOURCE DEVOLOPMENT

(Government of Kerala undertaking)

PROJECT REPORT

ON

CASN-CAMPUS_CONNECT

SUBMITTED BY

CHINMAYA V V

AY22CCSR13

DEPARTMENT OF COMPUTER SCIENCE

2024-2025

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CERTIFICATE

This is to certify that is a Bonafide record of the original work done by **CHINMAYA V V** Reg. No. **AY22CCSR13** of **COLLEGE OF APPLIED SCIENCE PAYYANUR,NERUVAMBRAM, KANNUR UNIVERSITY** for the partial fulfillment of the completion of Sixth semester **BSc. COMPUTER SCIENCE**

Head of the Department Principal

External Examiner Internal Examiner

1.

2.

DECLARATION

I CHINMAYA V V sixth semester BSc. Computer Science student of College of Applied Science Payyannur, Neruvambram under Kannur University do hereby declare that the project entitled CASN-CAMPUS_CONNECT: A complete solution for college staff, students and public is an original work done by we during the period of study at College of Applied Science Payyannur, Neruvambram under the supervision of Ms. Sneha, faculty member of Collage of Applied Science Payyannur, Neruvambrum.

Place: Neruvambram

Sincerely
CHINMAYA V V

ACKNOWLEDGEMENT

The successful completion of our project required a lot of guidance and assistance from many people, and we are extremely grateful to have received their support. We would like to express our sincere thanks to Principal Prof. Jaison D Joseph of the College of Applied Science Payyannur, Neruvambram, for providing the necessary facilities. We also extend our gratitude to Ms. Sneha, faculty of the Computer Science department, who took a keen interest in our project and guided us throughout the entire process by providing us with all the necessary information to develop a good system. We owe our success to their supervision and assistance, and we will never forget to thank them for their invaluable contributions to our project.

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1. ABSTRACT

Our proposed project entails the creation of a comprehensive digital platform designed to streamline information access and support for students, faculty, and visitors. This user-centric portal provides details on academic details, faculty profiles, and other essential resources to support the needs of the college community.

A key innovation is the AI-powered chatbot, developed to offer assistance for college-related inquiries. This chatbot can answer questions on topics. By offering instant, eligible responses, the chatbot enhances user engagement and makes essential information readily accessible, anytime and anywhere.

This platform reflects IHRD Neruvambram's commitment to innovation and student support, empowering users with a seamless, interactive experience and fostering a well-informed academic environment.

2. INTRODUCTION

Our project aims to create a digital platform designed to make information easily accessible for students, faculty, and visitors at IHRD Neruvambram. This platform will provide important details like academic information, faculty profiles, and other resources to meet the needs of the college community.

A key feature of the platform is an AI-powered chatbot that can answer questions related to the college. This chatbot will offer quick and reliable responses, helping users get the information they need anytime, anywhere.

This platform reflects IHRD Neruvambram's commitment to innovation and student support, providing a user-friendly, interactive experience to create a more connected and informed academic environment with a defined scope to:

Project Scope:

➢ **Key Features:**

- **Information Hub:**
  - Academic details such as course information, schedules, exam timelines, and results.
  - Faculty profiles including qualifications, courses taught, and contact details.
  - Additional resources like event updates, campus services, and news.

- **AI-powered Chatbot:**
  - Offers real-time, accurate responses to user queries.
  - Provides information on academic schedules, college events, faculty, admission details, etc.
  - Available 24/7 to assist with immediate needs, reducing wait times for information.

- **User-friendly Interface:**
  - Accessible on both mobile and desktop devices with a responsive design.
  - Simple, intuitive navigation for all users, regardless of tech proficiency.

- **Data Security & Privacy:**

  - Secure login system to ensure the protection of personal and academic data.
  - Role-based access control to restrict access to sensitive information.

➢ **Motivating the Need:**

The need for this platform arises from the challenges that students, faculty, and visitors currently face in accessing timely and accurate information. At present, the college community relies on various traditional channels like notice boards, emails, and in-person inquiries, which can be time-consuming and inefficient. This project aims to bridge that gap by creating a centralized, digital solution where all the relevant information is just a few clicks away.

Moreover, the introduction of an AI-powered chatbot helps to address common issues like long waiting times for responses and the inconsistency of information provided across multiple platforms. With the chatbot being available 24/7, it will provide immediate answers to user queries, further enhancing the platform's value.

The college’s commitment to embracing technology and enhancing the student experience is a core motivator for this initiative. By developing this platform, IHRD Neruvambram will not only streamline communication but also position itself as a forward-thinking institution in line with modern educational needs.

- **Project Vision:**

The vision of this project is to create a connected, efficient, and informed academic environment at IHRD Neruvambram. The platform will empower students, faculty, and visitors to easily access crucial information, facilitating their academic and administrative needs with ease. The AI-powered chatbot will serve as a cornerstone of the user experience, offering real-time, accurate responses to inquiries and supporting the platform's overall functionality.

By developing this platform, IHRD Neruvambram will reinforce its commitment to innovation, making education more accessible and transparent while enhancing student support. The ultimate goal is to foster a more engaged, informed, and efficient community, where everyone, regardless of their role, has quick access to the resources they need to succeed.

3. SYSTEM ANALYSIS

System analysis is the process of gathering and interpreting facts, diagnosing problems and using the facts to improve the system. Analysis is a detailed study of various operation performed by a system and their relationship within and outside of the system. This involves gathering information and using structured tools for analysis.

3.1 EXISTING SYSTEM

The current college information system may rely heavily on traditional methods such as phone calls, in-person inquiries, and a basic website with static content. While this provides essential information, it lacks interactivity, personalization, and real-time engagement. As a result, students and visitors often experience delays when seeking information, especially after office hours. The absence of automated responses also means that users may find it challenging to obtain specific answers, potentially leading to information gaps and dissatisfaction.

3.2 DRAWBACKS OF THE EXISTING SYSTEM

The existing system has some limitations are as follows

- Limited accessibility to information outside office hours.
- Lack of automated assistance for asked questions.
- Static information updates with minimal user interactivity.
- Heavy dependence on manual resources for responding to inquiries.

3.3 PROPOSED SYSTEM

To modernize the way information is shared and accessed, IHRD Neruvambram, Kannur, is introducing an enhanced college website integrated with a responsive AI-powered chatbot. This new system is set to address the limitations of traditional information access, such as limited office hours and manual handling of inquiries, by providing instant responses to frequently asked questions regarding admissions, academic programs, campus events, and more. The chatbot will be available 24/7, ensuring that users can receive assistance anytime, even after office hours, making it easier for prospective students and parents to connect with the college. This addition to the website reflects a commitment to improving user experience, reducing administrative workload, and fostering a more accessible, interactive platform for the college community. Through this initiative, IHRD Neruvambram aims to enhance engagement, streamline information delivery, and create a digital environment that supports both current and future students.

3.4 ADVANTAGE OF THE PROPOSED SYSTEM

- **Easy Access:** The platform brings all important information like academic details and faculty profiles in one place, making it easy to find what you need.

- **24/7 Support:** The AI chatbot provides instant help with college-related questions at any time of day or night.

- **User-Friendly:** The platform is designed to be simple and easy to navigate, making it easy for users to find information quickly.
- **Time-Saving:** The chatbot answers questions instantly, saving time for both students and faculty by reducing wait times.
- **Better Communication:** The platform improves communication by offering clear and accurate information, keeping everyone on the same page.
- **Supports Remote Access:** Students and faculty can use the platform from anywhere, making it easier to access information remotely.
- **Encourages Innovation:** The use of AI technology shows the college's commitment to staying up-to-date with digital trends and providing a modern experience.
- **Lowers Staff Workload:** The chatbot answers common questions, allowing staff to focus on more complex tasks

3.5 MODULES

1. ADMIN MODULE

This module mainly deals with manage staff, students, department, course and subject, etc..

2. STAFF MODULE

This module was added by the admin. This module mainly deals with manage material ,chat with parent and manage assignment notification and add attendance .

3. HOD MODULE

This module was added by admin. This deals with subject allocation to course, subject allocation to staff and class in charge allocation.

4. STUDENT MODULE

This module was added by admin. This module mainly deals with submit assignment.

5. PARENT MODULE

This module mainly deals with the registration of parent, chat with staff, view attendance of children and view meetings.

6. PUBLIC MODULE

This module mainly deals with asking questions to chatbot , view courses, view previous year question paper and view faculties.

4. FEASIBILITY STUDY

A feasibility analysis usually involves a thorough assessment of the operational (need),financial and technical aspects of a proposal. Feasibility study is the test of the system proposal made to identify whether the user needs may be satisfied using the current software and hardware technologies, whether the system will be cost effective from a business point of view and whether it can be developed with the given budgetary constraints. A feasibility study should be relatively cheap and done at the earliest possible time. Depending on the study, the decision is made whether to go ahead with a more detailed analysis. All projects are feasible when given unlimited resources and infinite time. It is both necessary and prudent to evaluate the feasibility of a project at the earliest possible time. A feasibility study is not warranted for systems in which economic justification is obvious, technical risk is low, few legal problems are expected and no reasonable alternative exists. An estimate is made of whether the identified user needs may be satisfied using current software and hardware technologies.

The study will decide if the proposed system will be cost effective from the business point of view and if it can be developed in the given existing budgetary constraints. The feasibility study should be relatively cheap and quick. The result should inform the decision of whether to go ahead with a more detailed analysis.

Feasibility study may be documented as a separated report to higher officials of the top-level management and can be include as an appendix to the system specification. Feasibility and risk analysis is related in many ways. If there is more project risk then the feasibility of producing the quality software is reduced.

When a new project is proposed, it normally goes through feasibility assessment. Feasibility study is carried out to determine whether the proposed system is possible to develop with available resources and what should be the cost consideration. Facts considered in the feasibility analysis were.

- Operational feasibility
- Technical feasibility
- Economical feasibility
- Behavioral feasibility
- Hardware feasibility
- Software feasibility

OPERATIONAL FEASIBILITY

Operational feasibility is used for identify the importance of certain problem in project and how it is to be solved. This also measures how solution of the problems will work for any project or in any organization .It analysis the behavior of the proposed system and whether the proposed system is easier than the existing system for the users of the system. The proposed project is operationally feasible. Because the proposed android application is easily operated by the user. The user who have basic computer knowledge, he can use the application for registration and then use the application for their intended applications. For operating the android application, the user need only the basic internet knowledge. Hence the proposed system is operationally feasible.

TECHNICAL FEASIBILITY

Technical feasibility is used to identify whether the technical resources are available to form the projector system. It suggests efficient input and output devices to manage large amount of data. It also measure the hardware and software of existing system to which extent it can support the proposed system. It checks the available technology is with in the given constraints such as budget. The proposed system meets all the requirement of the Technical feasibility. Because the implementation of the project need no technological difficult. The web part of the project can be easily implemented in user friendly IDE. The android part of the project can be easily implemented in a user friendly IDE, that is android studio. So the Development of the project does not meet any type of the technical difficult. Hence we can be say the proposed system is technically feasible.

ECONOMIC FEASIBILITY

Economic feasibility measures all cost incurred in development of new system. It is used to determine the financial resources of the project. It is called cost benefit analysis because it the determines the total cost for development of new system and benefits derived from the new system. The proposed project is economically feasible. Because once the system is put in to its use in the current market the system provides economical advantage to the firm. Also the firm can 12 afford the cost to implement the project. The proposed project provide tangible and intangible benefits comparing to the existing projects. This system does not need any initial investment and it can improve the quality of service.

BEHAVIOURAL FEASIBILITY

It evaluates and estimates the user attitude or behaviour towards the development of new system. It helps in determining if the system requires special effort to educate, retrain, transfer and changes in employees job status on new ways of conducting business. The proposed system has a high operational feasibility. Because the application is accessible at anywhere and anytime. For operating the application, we only need the basic android knowledge. The changes in the hardware or software environment does not affect the system. The people are inherently change the computer. But the changes in the does not affect the system.

SOFTWARE FEASIBILITY

Even though software is developed in a very high software environment, it will be supported by many other platform and environment with minimum changes.

HARDWARE FEASIBILITY

The software can be developed with resource already existing. Here the consideration is that the existing hardware resources support the technologies that are to be used by the new system. No hardware was newly bought for the project and hence. Software is said to achieve hardware feasibility**.**

5. SYSTEM REQUIREMENTS

5.1 HARDWARE REQUIREMENTS

- PROESSOR : Intel Pentium Core i3 and above
- MEMORY(RAM) : 4GB RAM and above
- STORAGE : 320 GB hard disk and above

5.2 SOFTWARE REQUIREMENTS

One of the most difficult tasks is selecting software for the system, once the system requirements is found out then we have to determine whether a particular software package fits for those system requirements. The application requirement:

- FRONTEND : HTML ,CSS ,BOOTSTRAP , JS , XML,JAVA,PYTHON
- FRAMEWORK : Django
- IDE : PyCharm , Adobe Dream weaver , SQL yog , Android studio
- BACKEND : MySQL ,PYTHON
- ANDROID OS :Android M or above
- OPERATING SYSTEM : Windows 8 or above

5.3 SOFTWARE FEATURES

Windows 8

Windows 8, launched by Microsoft on October 26, 2012, marked a significant shift in the way Windows operated, focusing heavily on both desktop and tablet experiences. It was the successor to Windows 7 and was designed with a dual-purpose interface to work well on both traditional PCs and touch-enabled devices like tablets.
Key features of Windows 8 included:

1. **Start Screen**: Replacing the classic Start Menu, the Start Screen was a grid of live tiles designed for touchscreens, offering quick access to apps and notifications.

2. **Windows Store**: A new app marketplace, similar to mobile app stores, allowing users to download and install modern-style apps.
3. **Touch Interface**: Optimized for touchscreen devices with features like swipe gestures, making it easier to interact on tablets and touch-enabled laptops.
4. **Fast Boot and Improved Performance**: Windows 8 provided faster boot times, better performance, and improved system security with features like Secure Boot and Windows Defender.
5. **Charms Bar**: A menu that could be accessed from the right edge of the screen, providing quick access to search, settings, and devices.

   Despite these innovations, Windows 8 received mixed feedback. The shift away from the familiar Start Menu and the emphasis on the new touch interface led to criticism, particularly among traditional desktop users. This led Microsoft to release Windows 8.1 in 2013, which reintroduced a more familiar Start button and improved usability. In the end, Windows 8 set the stage for Windows 10, which aimed to balance the needs of both touch and desktop users and became more widely accepted.

HTML (HyperText Markup Language)

HTML, standing as the cornerstone of web development, provides the fundamental structure and content of web pages. Through a system of tags and elements, HTML enables developers to define the layout, hierarchy, and semantic meaning of the content displayed on a webpage. From headings and paragraphs to images, links, and forms, HTML offers a versatile set of tools for organizing and presenting information. Its simplicity and universality make it accessible to beginners while providing the flexibility and extensibility required for complex web applications. As the backbone of the web, HTML forms the foundation upon which CSS and JavaScript are layered, ensuring a well-structured and semantically meaningful browsing experience for users worldwide.

CSS (Cascading Style Sheets)

CSS, often referred to as the design language of the web, empowers developers to control the visual presentation and layout of web pages. By defining styles for HTML elements, CSS enables designers to customize colors, fonts, spacing, and other aesthetic properties, creating visually appealing and cohesive user interfaces. Through the use of selectors, properties, and values, CSS provides a flexible and efficient

means of styling web content, facilitating responsive design and cross-browser compatibility. Its separation from HTML promotes modularization and reusability, allowing developers to maintain consistent branding and design across multiple pages or applications. With CSS, designers can unleash their creativity, transforming static HTML structures into dynamic and engaging user experiences that captivate and delight visitors.

BOOTSTRAP

Bootstrap is an open-source framework that helps web developers build websites quickly and easily. It was created by Twitter developers Mark Otto and Jacob Thornton in 2011. The goal of Bootstrap is to make designing responsive, mobile-friendly websites simple, with a consistent and professional look.

One of the main features of Bootstrap is its responsive design. With more people using smartphones and tablets, having websites that adjust to different screen sizes is crucial. Bootstrap's grid system divides the screen into a flexible 12-column layout, making it easier to organize content in rows and columns that look good on any device.

Bootstrap also comes with many pre-designed components, like buttons, navigation bars, and forms. These ready-to-use elements save developers time and ensure a consistent design across the website. Additionally, Bootstrap includes JavaScript plugins for adding interactive features, such as carousels, modals, and dropdowns, which improve user experience.

The framework is compatible with all major web browsers, ensuring that websites created with Bootstrap look the same no matter which browser someone is using. This makes it easier for developers to create cross-browser websites without extra effort.

Since Bootstrap is open-source, it is free to use, and its large community of developers constantly updates it with new features and fixes. This means developers have access to great support and can easily find solutions to any challenges they face.

In conclusion, Bootstrap is a powerful tool for web development, making it easier to create responsive, user-friendly, and visually appealing websites. Its simplicity, flexibility, and strong community support make it one of the most popular frameworks among web developers today.

JavaScript

JavaScript, heralded as the programming language of the web, empowers developers to add interactivity and dynamic behavior to web pages. As a versatile and lightweight language, JavaScript enables the creation of interactive elements, animations, and real-time updates that engage users and enhance the browsing experience. With support for event-driven programming and asynchronous operations, JavaScript facilitates responsive and interactive web applications, enabling seamless interactions with users and external data sources. Its extensive ecosystem of libraries and frameworks, such as React, Angular, and Vue.js, further extends its capabilities, enabling developers to build complex single-page applications (SPAs) and progressive web apps (PWAs) with ease. From simple form validation to complex data visualizations, JavaScript remains a vital tool in the modern web developer's arsenal, empowering them to create immersive and dynamic experiences that push the boundaries of what's possible on the web.

Python and Django

Python, renowned for its simplicity, readability, and versatility, stands as one of the most popular programming languages globally. With its clear and concise syntax, Python facilitates rapid development and fosters a collaborative and inclusive coding environment. As a high-level language, Python abstracts complex programming concepts, making it accessible to beginners while providing advanced features and libraries for seasoned developers. Its extensive standard library and vibrant ecosystem of third-party packages enable developers to tackle a wide range of tasks, from web development and data analysis to machine learning and automation.

Django, a high-level Python web framework, empowers developers to build robust and scalable web applications with ease. Built on the principles of DRY (Don't Repeat Yourself) and convention over configuration, Django streamlines the development process, allowing developers to focus on building features rather than boilerplate code. With built-in components for authentication, authorization, routing, templating, and database management, Django provides a comprehensive toolkit for building secure and maintainable web applications. Its emphasis on security, scalability, and performance makes it a popular choice for organizations of all sizes, from startups to Fortune 500 companies.

Together, Python and Django form a powerful combination for web development, enabling developers to create elegant and feature-rich applications that meet the demands of today's digital landscape. With Python's simplicity and versatility complemented by Django's robustness and scalability, developers can unleash their creativity and build web applications that push the boundaries of what's possible. Whether it's building a simple blog or a complex enterprise-level application, Python and Django empower developers to turn their ideas into reality and deliver exceptional user experiences on the web.

Advantages of Python and Django

Python and Django offer a myriad of advantages for web development projects. Python's simplicity, readability, and extensive library ecosystem make it a favorite among developers for its ease of learning and rapid development. Its versatility extends from web development to data science, machine learning, and automation, providing a unified language for diverse project requirements. Meanwhile, Django's high-level framework streamlines development with its DRY (Don't Repeat Yourself) philosophy and batteries-included approach. Its built-in features for authentication, database management, and URL routing reduce development time and minimize boilerplate code. Additionally, Django's emphasis on security, scalability, and performance ensures robust and stable web applications suitable for enterprise-level deployments. Together, Python and Django empower developers to build sophisticated and scalable web applications efficiently, enabling them to focus on crafting features and delivering exceptional user experiences without compromising on security or performance.

MySQL

MySQL is an open-source relational database management system (RDBMS) that uses Structured Query Language (SQL) to manage and manipulate data. Initially created by MySQL AB in 1995 and later acquired by Oracle Corporation, MySQL is widely used for managing databases in various applications, especially web-based systems.

Key Features of MySQL:

1. **Relational Database:** MySQL stores data in tables, and relationships can be defined between tables using keys (primary, foreign).
2. **SQL-based:** MySQL uses SQL to query and manage the data, offering a standard way to interact with databases.
3. **Open-source and Free:** MySQL is open-source, meaning it is free to use and modify. This makes it highly popular among developers.
4. **Cross-platform Support:** MySQL works on multiple operating systems, including Linux, Windows, and macOS, making it flexible and versatile.
5. **High Performance and Scalability:** Known for its speed and ability to handle large amounts of data, MySQL is suitable for both small and large applications, from personal projects to enterprise-level systems.
6. **Security Features:** MySQL provides robust security measures such as user authentication, access control, and data encryption to safeguard sensitive information.
7. **Widely Used in Web Development:** MySQL is commonly used with programming languages such as PHP and Python in building dynamic, data-driven websites and applications, particularly in the LAMP (Linux, Apache, MySQL, PHP) stack.
8. **Active Community:** Being open-source, MySQL has a large and active community, offering ongoing support, frequent updates, and extensive documentation.

In summary, MySQL is a powerful, reliable, and efficient RDBMS that continues to be one of the most popular database management systems worldwide, particularly for web-based applications. Its flexibility, scalability, and strong community support make it a go-to choice for developers and businesses alike.

6. DESIGN

6.1 DATA FLOW DIAGRAMS(DFD)

A data flow diagram (DFD) is a graphical representation of the "flow" of data through an information system. Data flow diagrams are used to describe how the system transforms information. They define how information is processed and stored and identify how the information flows through the processes. DFDs can also be used for the visualization of data processing (structured design).

With a data flow diagram, users are able to visualize how the system will operate, what the system will accomplishes, and how the system will be implemented. The old system's data flow diagrams can be drawn up and compared with the new system's data flow diagrams to draw comparisons to implement a more efficient system. Data flow diagram can be used to provide the end user with a physical idea of where the data they input ultimately has an effect upon the structure of the whole system from order to dispatch to report. How any system is developed can be determined through a data flow diagram.

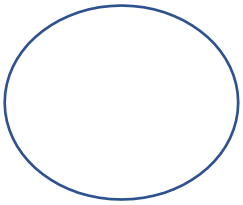
A DFD provides no information about the timing of processes, or about whether processes will operate in sequence or in parallel. It is therefore quite different from a flowchart. Which shows the flow of control through an algorithm, allowing a reader to determine what operation will be performed, in what order, and under what circumstances, but not what kinds of data will be input to and output from the system, nor where the data will come from and go to, nor where the data will be stored (all of which are shown on a DFD).

DFD Symbols

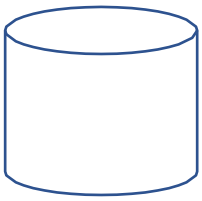
In the DFD, there are four types of symbols

Represents data source or destination

Represent the flow of data



Represent a process that transforms the data



Represent Data storage

6.2 RULES USED IN CONSTRUCTING DFD

• Process should be named and numbered

• The direction of flows top to bottom & from left to right after exploding, lower level details of process are to be numbered.

• The name of data stores, sources, destinations are written in upper cases process and data flow names have first letters of each word capitalized.

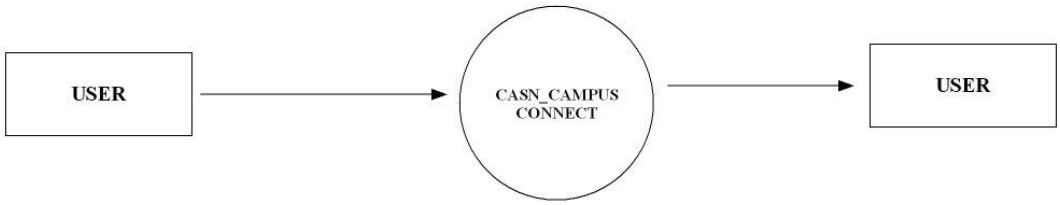
It is common practice to draw a context level data flow diagram first, which shows the interaction between the system and external agents which act as data sources and data links. On the context diagram (also known as the level 0 DFD') the system's interactions with the outside world are modeled purely in terms of data flows across the system's boundary. The context diagram shows the entire system as a single process, and gives no clues as to its internal organization. This context level DFD is next "exploded", to produce a level 1 DFD that shows some of the detail of the system being modeled. The level 1 DFD shows how the systems is divided into sub systems, each of which deals with one or more of the data flows to or from an external agent, and which together

provide all of the functionality of the system as a whole. It also identifies internal data stores that must be present in order for the system to do its jobs, and shows the flow of data between the various parts of the systems.

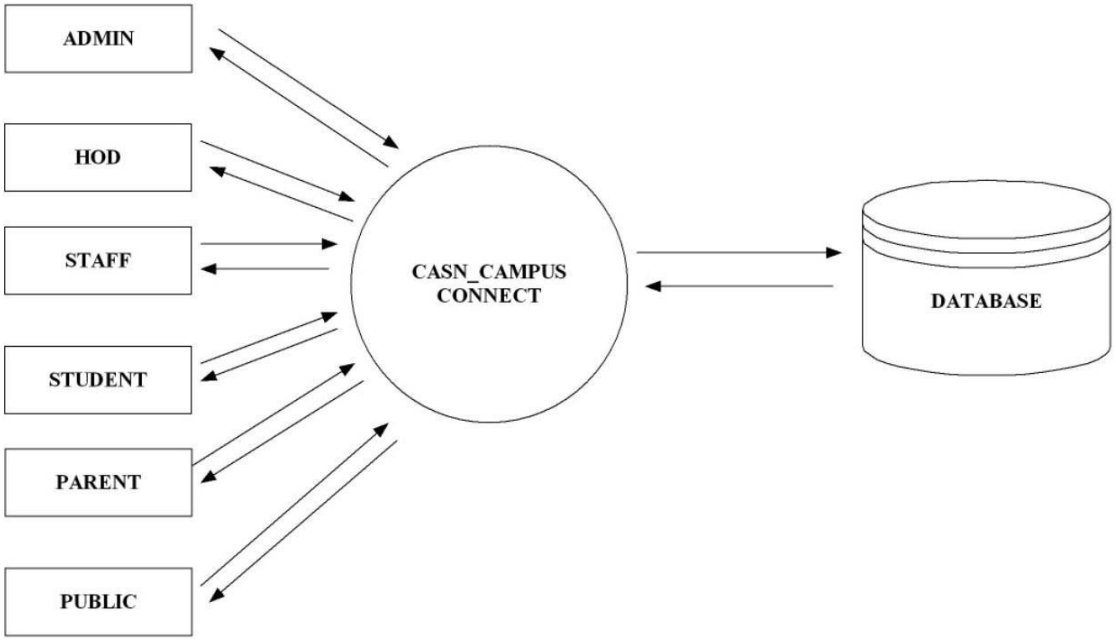
Data flow diagrams are constructed with four commonly encountered data flow diagram symbols they are, the circle or bubble represents a transformation process, and the label inside the bubble describes the process using an active verb to do so. Data flows are directed lines that identify the input data flows and output data flows at each process bubble. Data storage is represented by an open ended rectangle with a label that identifies the data store, or file. The square is labeled to identify an external entity that is a source or destination of data flow.

6.3 DATA FLOW DIAGRAMS OF CASN_CAMPUS CONNECT

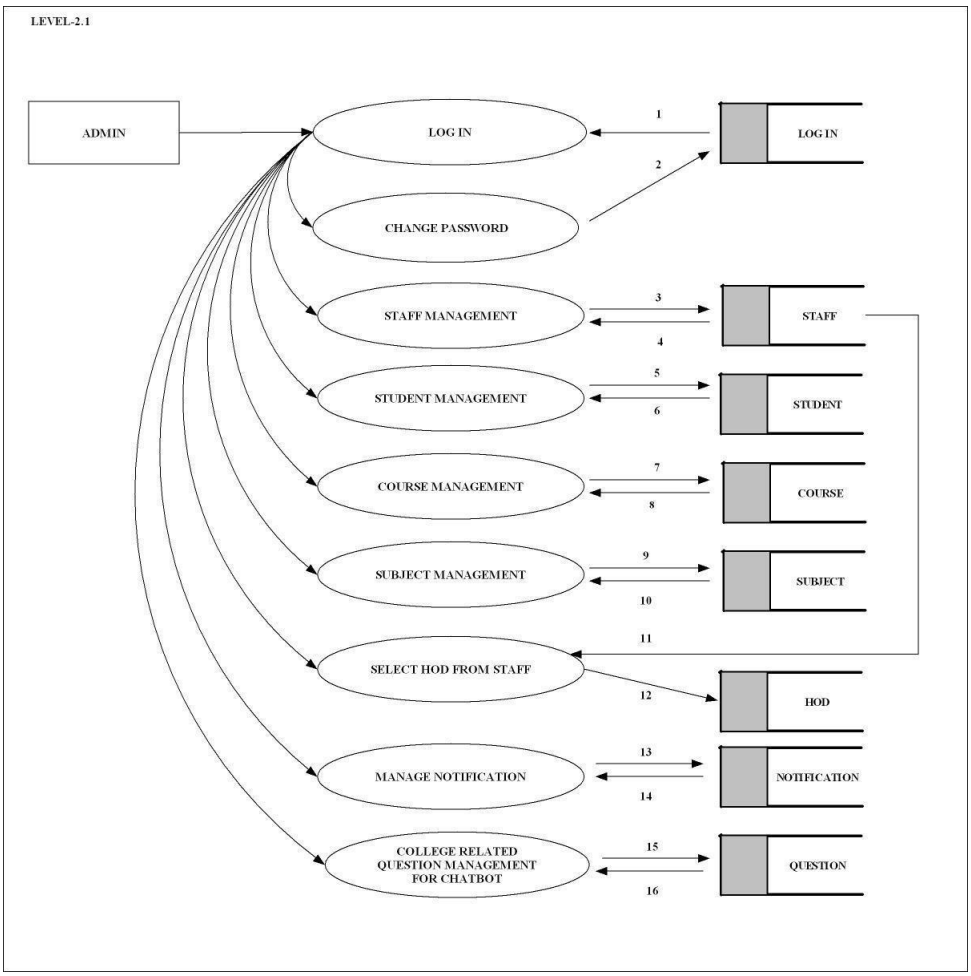
LEVEL 0



LEVEL 1

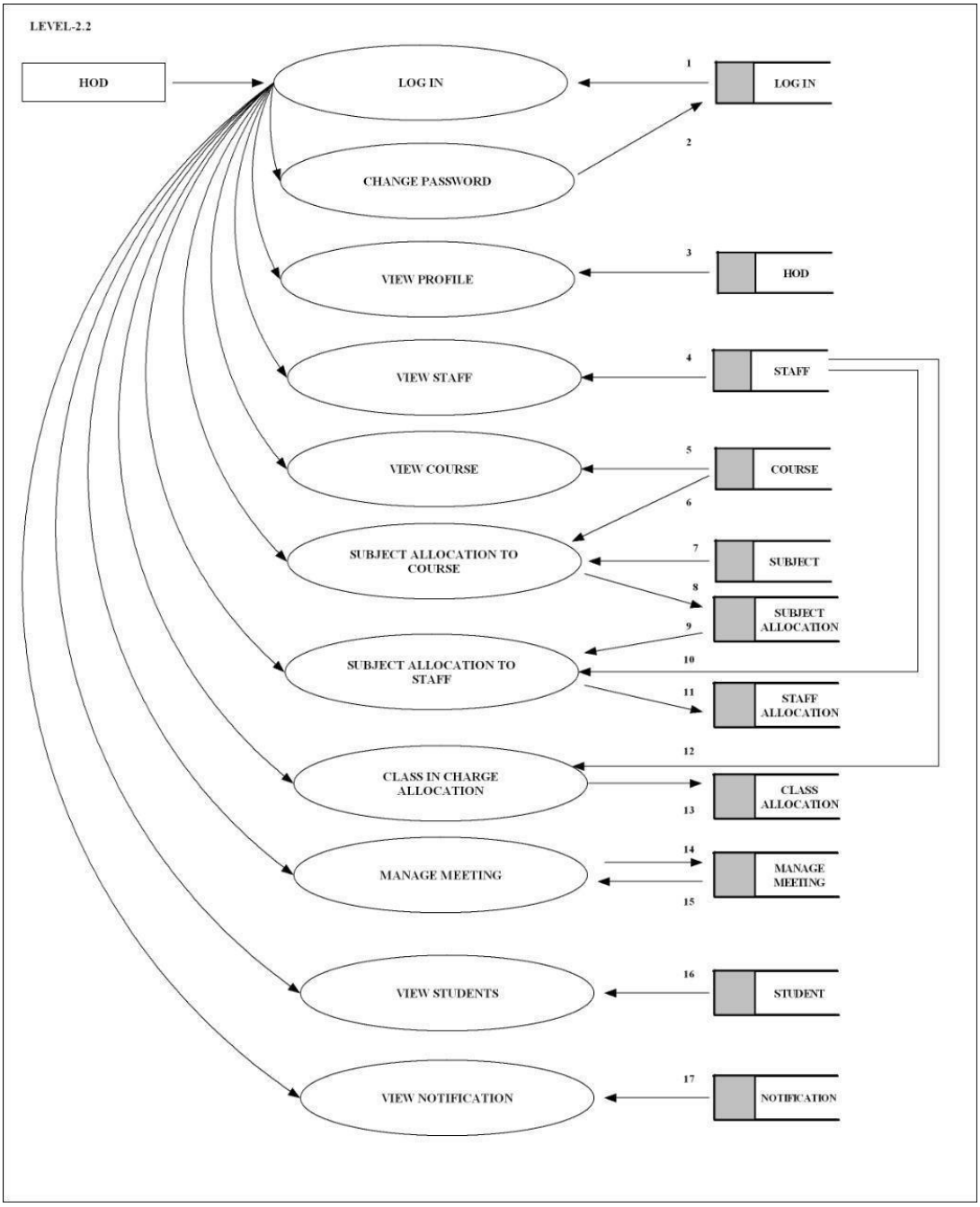


LEVEL 1.1 ADMIN



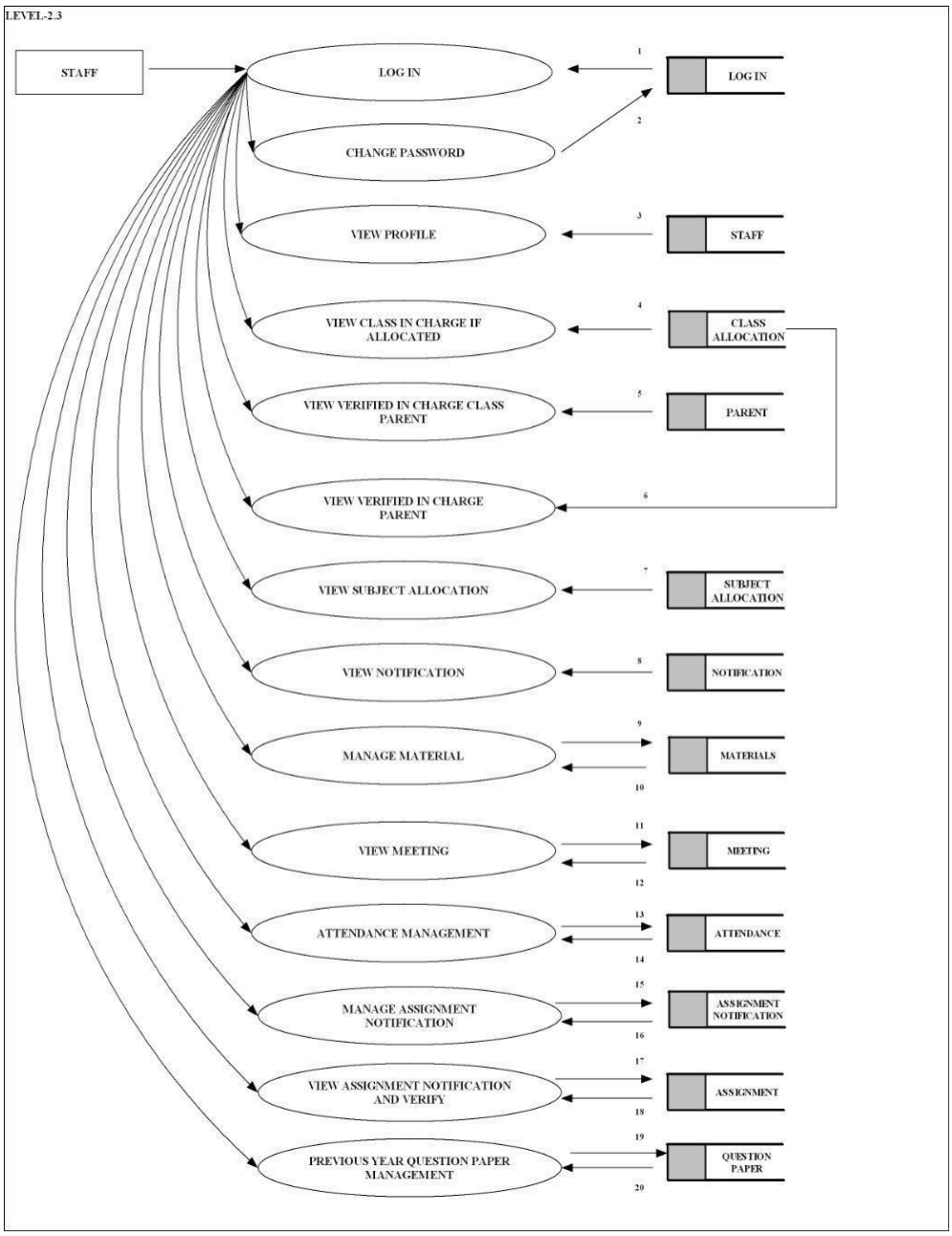
1. Login details
2. Login details
3. Staff details
4. Staff details
5. Student details
6. Student details
7. Course details
8. Course details
9. Subject details
10. Subject details
11. Staff details
12. Hod details
13. Notification details
14. Notification details
15. Question details
16. Question details

LEVEL 1.2 HOD


LEVEL-2.2
HOD
LOG IN
CHANGE PASSWORD
VIEW PROFILE
VIEW STAFF
VIEW COURSE
SUBJECT ALLOCATION TO COURSE
SUBJECT ALLOCATION TO STAFF
CLASS IN CHARGE ALLOCATION
MANAGE MEETING
VIEW STUDENTS
VIEW NOTIFICATION
LOG IN
HOD
STAFF
COURSE
SUBJECT
SUBJECT ALLOCATION
STAFF ALLOCATION
CLASS ALLOCATION
MANAGE MEETING
STUDENT
NOTIFICATION
1
2
3
4
5
6
7
8
9
10
11
12
13
14
15
16
17

1. Login details
2. Login details
3. Hod details
4. Staff details
5. Course details
6. Course details
7. Subject details
8. Subject allocation details
9. Subject allocation details
10. Staff details
11. Staff allocation details
12. Staff details
13. Class allocation details
14. Manage meeting details
15. Manage meeting details
16. Student details
17. Notification details

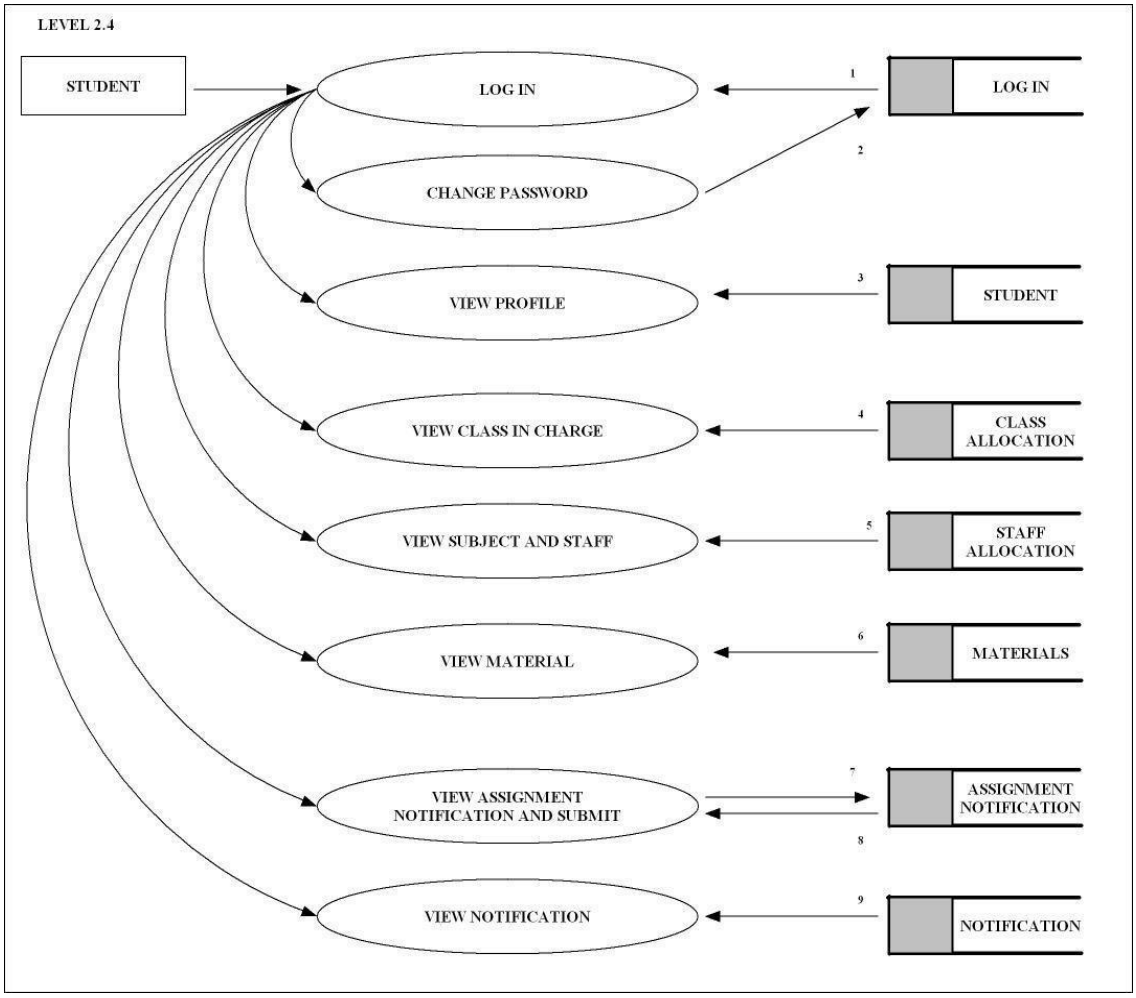
LEVEL 1.3 STAFF



1. Login details
2. Login details
3. Staff details

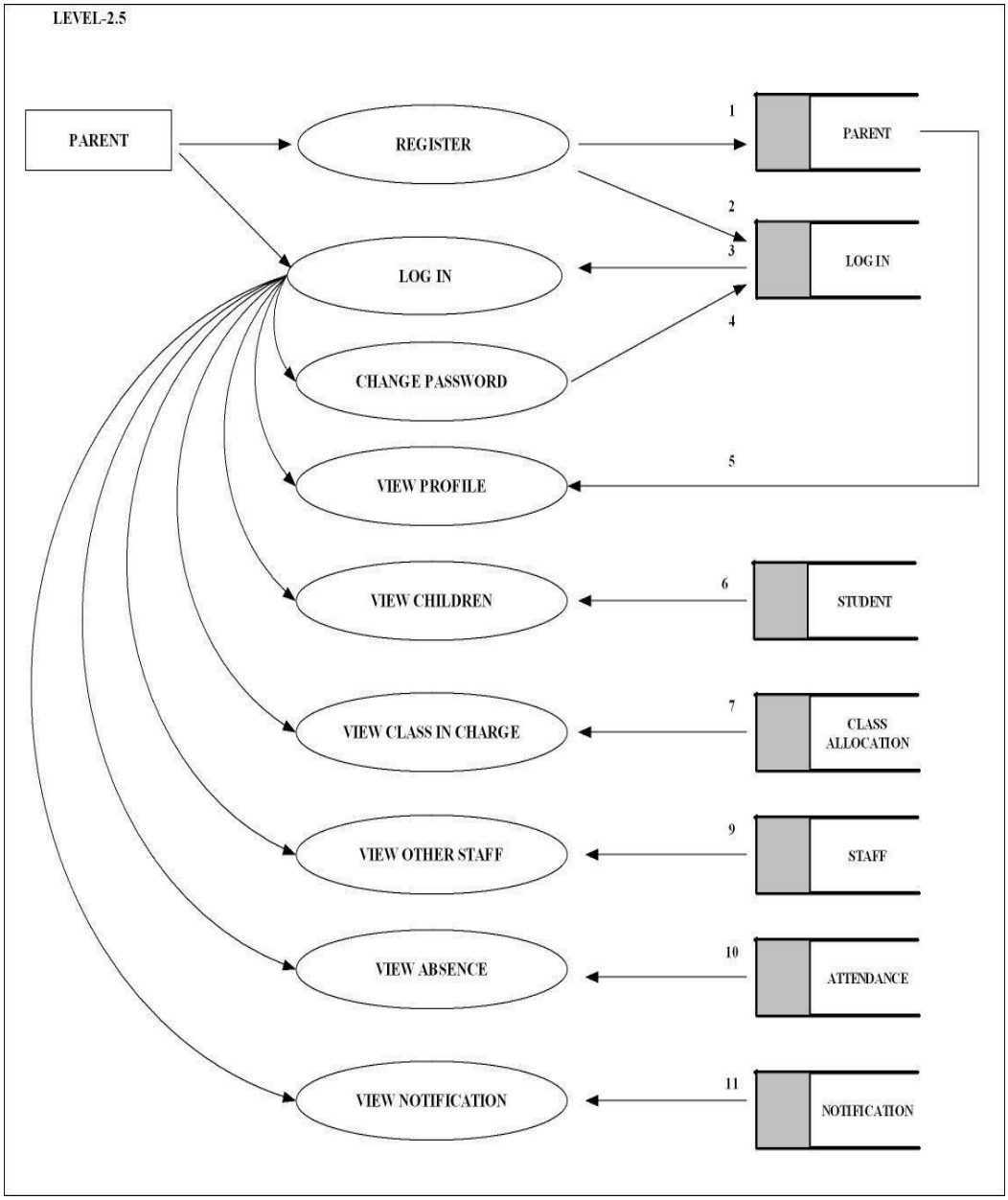
4. Class allocation details
5. Parent details
6. Class allocation details
7. Subject allocation details
8. Notification details
9. Materials details
10. Materials details
11. Meeting details
12. Meeting details
13. Attendance details
14. Attendance details
15. Assignment notification details
16. Assignment notification details
17. Assignment details
18. Assignment details
19. Question paper details
20. Question paper details

LEVEL 1.4 STUDENT



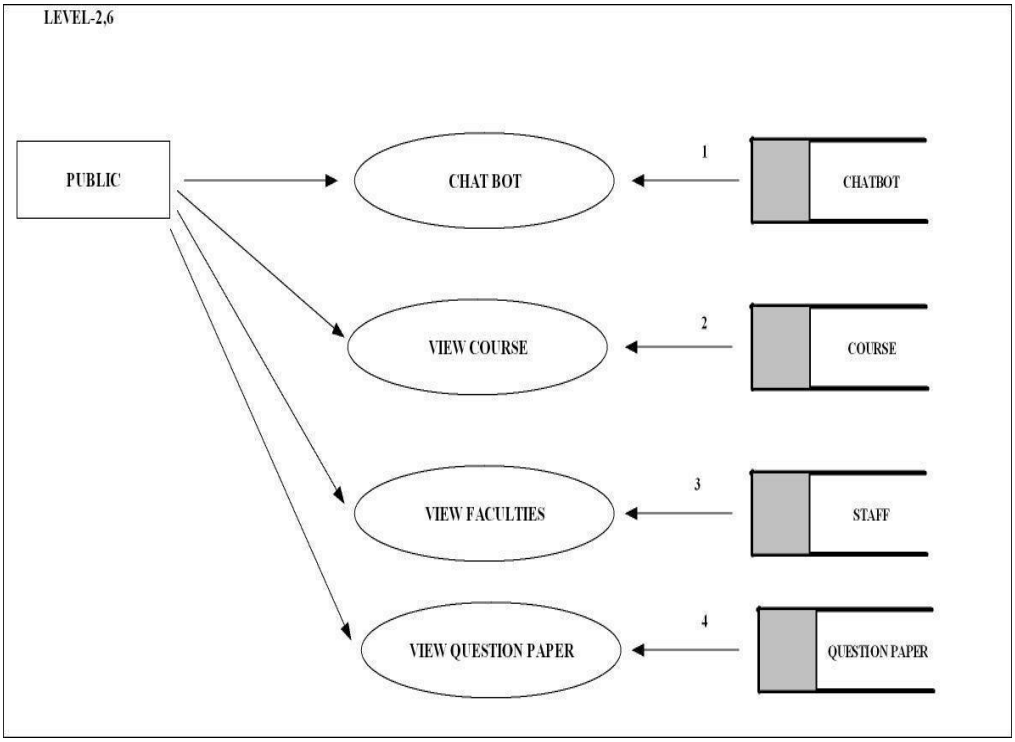
1. Login details
2. Login details
3. Student details
4. Class allocation details
5. Staff allocation details
6. Materials details
7. Assignment notification details
8. Assignment notification details
9. Notifications details

LEVEL 1.5 PARENT



1. Parent details
2. Login details
3. Login details
4. Login details
5. Parent details
6. Student details
7. Class allocation details
8. Staff details
9. Attendance details
10. Notification details

LEVEL 1.6 PUBLIC



1. Chat bot details
2. Courses details
3. Staff details
4. Question paper details

6.4 DATABASE DESIGN

A Data base management system provides flexibility in the storage and retrieval of data and production of information. The various data elements and their relationships and data structures have been identified in the analysis stage. The starting point for the database design in the dictionary developed in the analysis phase. Data base design is a two phase process. In the first phase the logical database design is made, where a conceptual data structure is built, based on the users requirements. The second phase is called database design. It is the process of formulating the logical design in terms of database management system facilities. Thus a data base is a connection of logically related records. The necessary tables and their fields are identified in the phase. The design should be done in a way the information stored in the data base can retrieve quickly whenever necessary.

The primary functions of data base management systems are

- Creation of data base.
- Data access and manipulation.
- Retrieval of data.
- Data Base Management System provides integrity and security

Input design

Input design is the process of converting user-oriented inputs to computer -based format. The input to the system can be defined as the information that is used for processing by the system to obtain meaningful information, which helps in decision-making input design is the process of converting user- oriented inputs to a computer based format. Input design is the most expensive phase of the project and may cause some major problems if not designed with care. The quality of system is determined from the quality of system outputs which in turn is determined from input.

Input design features can ensure the reliability of the system and also determines how efficiently the user can interact with the system. A good input screen is clear and

free from irrelevant information. Entered data has to be checked for their accuracy and detection of errors. All the data entry screens should be of interactive nature so that the user can directly input data according to prompt messages. The input design determines whether the user can interact directly with the computer. Interactive input screens ensure the reliability and accuracy of the system.

All the input screens used here have script validation for null checking and data types wherever necessary. All the programs inserting data to the table checks the length of the values in addition to the data type verification. They also prompt they entry of values in mandatory fields. The goal of designing input data is to make it free from logical errors. The input data is also used for easy calculation of necessary functions. Different forms are used for data screens in order to input data into the system. The screen formats have options like add, delete, edit and update with buttons, which is used for easy input and retrieving information. All the fields are validated.
If the user enters invalid data, appropriate messages are displayed. Input specification describes the manner in which data enter the system for processing.

Output design

One of the most important features of system for users is the output is produces. Output design should improve the system relationship with the user it produces and help in decision making. Considering the future use of the output required and depending on the nature, they displayed on the monitor immediate need or obtain the hard copy. Computer output is the most important and direct source of information to the user. Output needs to be in an attractive manner. Efficient, intelligent output design improves the system relationship between the users and helps in decision making.

The objective of output design is to define the control and formats of all printed, documented, reports and screens that will be produced by the system. Computer output is the most important and direct source of developing the system and the basis on which they will evaluate the usefulness of the application. Output generally refers to the results that are generated by the system is designed so as to include a number of reports. Reports reflect the output design.

Objectives of the output design are:

- Design inputs to serve the indented purpose
- Design output to fit the user
- Deliver the appropriate quantity of output
- Assure that output where it is needed
- Provide output on time
- Choose the right output method

6.5 NORMALISATION

Normalization is the process of efficiently organizing data in a database. It is a systematic way of ensuring that a database structure is suitable for general purpose querying and free of certain undesirable characteristics insertion, update, and deletion anomalies that could lead a loss of data integrity. Edgar F.Codd, the inventor of the relational model, introduces the concept of normalization and what we now know as the First Normal form (INF) in 1970, Informally, a relational data base table (the computerized representation of a relation) is often described as 'normalized' if it is in the Third Normal Form. Most 3NF tables are free of insertion, update, and deletion anomalies that are in most cases 3NF table adhere to 4NF and SNF but typically not 6NF).

There are two goals of the normalization process: eliminating redundant data(for example, storing the same data in more than one table) and ensuring data dependencies make sense (only storing related data in a table). Both of these are worthy goals as they reduce the amount of space a database consumes and ensure that data is logically stored. All of this is done according to rules designed both to protect the data and to make the database more flexible. Data normalization is primarily used in transactional processing. The two key goals of the normalization process are eliminating redundant data and ensuring that data dependencies make sense. Eliminating redundant data implies storing the same data in more than one table and ensuring that data dependencies makes sense implies only storing related data in a table.

Putting these key goals together facilities reducing the amount of space a database consumes and also ensures that data is logically stored. Normalization is the way that data is grouped in the simplest way possible, so that later changes can be made with minimum impact on the data structure. Normalization helps the outcome of planning a database to be an efficient storage structure. And it should be pointed out that efficient doesn't necessarily size here.

FIRST NORMAL FORM:-

First normal form (INF) sets the very basic rules for an organized database.

These rules include

- Eliminate duplicative columns from the same table.
- Create separate tables for each group of related data and identify each row with a unique column or set of columns (the primary key).

SECOND NORMAL FORM:-

The second normal form says that each non key attribute should depend on a key. Second normal form (2NF) addresses the removal of duplicate data further, by first meeting all the requirements for first normal form. Also it removes subsets of data that apply to multiple rows of a table and places them in separate tables. Finally relationships are created between the new tables and their predecessors through the use of foreign keys.

THIRD NORMAL FORM:-

The third normal form says that non-key attributes should directly depend on a primary attribute. Third normal form (3NF) further addresses the structures of second normal form (2NF) by which are not dependent upon the primary key. These normalization guidelines are cumulative. For a database to be in 2NF, it must first fulfill all the criteria of a INF database.

6.6 TABLE DESIGN

The overall objective in the development of database technology has be to treat data as organizational resource and as a integrated whole. Data Base Management System allows data to protected and organized separately from other resources. Database design integrated collection of data. The most significant form of data as seen by the programmer is data as stored on the direct access storage device. This is the difference between logical and physical design.

Database is key source of information in to the system. The database should be properly designed planned or collection, accumulation, editing and retrieving the required information. This database contains tables where each table corresponds to one particular type of information. Each piece of information in the table is called a field or column. A table also contains records, which is set of fields with different information. There are primary key fields that uniquely identify record in the table.

TABLES

1. LOGIN TABLE

FIELD	TYPE	DESCRIPTION	CONSTRAINS
ID	Int	Login id	Primary key
username	Varchar	User name	Not null
password	Varchar	User password	Not null
usertype	Varchar	User type	Not null

2. STUDENT TABLE

FIELD	TYPE	DESCRIPTION	CONSTRAINS
ID	Int	Student id	Primary key
name	Varchar	Student name	Not null
place	Varchar	Student place	Not null
post	Varchar	Student post	Not null
pin	Varchar	Student pin	Not null
email	Varchar	Student email	Not null
phone	Varchar	Student phone	Not null
image	Varchar	Student image	Not null
LOGIN_id	Int	Login id	Not null, Foreignkey
Admissionnumber	Varchar	Student admission number	Not null
dateofbirth	Varchar	Student date of birth	Not null
COURSE_id	Int	Course id	Not null, Foreignkey
PARENT_id	Int	Parent id	Not null, Foreignkey
batch	Varchar	Student batch	Not null
semester	Varchar	Student semester	Not null

3. PARENT TABLE

FIELD	TYPE	DESCRIPTION	CONSTRAINS
ID	Int	Parent id	Primary key
name	Varchar	Parent name	Not null
phone	Varchar	Parent phone	Not null
email	Varchar	Parent email	Not null
LOGIN_id	Int	Login id	Not null, Foreignkey

4. STAFF TABLE

FIELD	TYPE	DESCRIPTION	CONSTRAINS
ID	Int	Staff id	Primary key
name	Varchar	Staff name	Not null
place	Varchar	Staff place	Not null
post	Varchar	Staff post	Not null
pin	Varchar	Staff pin	Not null
email	Varchar	Staff email	Not null
phone	Varchar	Staff phone	Not null
image	Varchar	Staff image	Not null
qualification	Varchar	Staff qualification	Not null
department	Varchar	Staff department	Not null
LOGIN_id	Int	Login id	Not null, Foreignkey

5. HOD TABLE

FIELD	TYPE	DESCRIPTION	CONSTRAINS
ID	Int	Hod id	Primary key
STAFF_id	Int	Staff id	Not null, Foreignkey
LOGIN_id	Int	Login id	Not null, Foreignkey

6. CHAT TABLE

FIELD	TYPE	DESCRIPTION	CONSTRAINS
ID	int	Chat id	Primary key
date	varchar	Chat date	Not null
type	varchar	Chat type	Not null
message	varchar	Message	Not null
STAFF_id	int	Staff id	Not null, Foreignkey
PARENT_id	int	Parent id	Not null, Foreignkey

3.

7. CHATBOT TABLE

FIELD	TYPE	DESCRIPTION	CONSTRAINS
ID	Int	Chatbot id	Primary key
type	Varchar	User type	Not null
email	Varchar	User email	Not null
message	Varchar	Message	Not null

8. SUBJECT TABLE

FIELD	TYPE	DESCRIPTION	CONSTRAINS
ID	Int	Subject id	Primary key
subjectname	Varchar	Subject name	Not null
syllabus	Varchar	Syllabus	Not null

9. NOTIFICATION TABLE

FIELD	TYPE	DESCRIPTION	CONSTRAINS
ID	Int	Notification id	Primary key
date	Varchar	Notification date	Not null
notification	Varchar	Notification	Not null

10. QUESTION TABLE

FIELD	TYPE	DESCRIPTION	CONSTRAINS
ID	Int	Question id	Primary key
question	Varchar	Questions	Not null
answer	Varchar	Answers	Not null

11. COURSE TABLE

FIELD	TYPE	DESCRIPTION	CONSTRAINS
ID	Int	Course id	Primary key
coursename	Varchar	Course name	Not null
department	Varchar	Department	Not null

12. STAFFALLOCATION TABLE

FIELD	TYPE	DESCRIPTION	CONSTRAINS
ID	Int	Staff allocation id	Primary key
STAFF_id	Int	Staff id	Not null
SUBJECTALLOCATION_id	Int	Subjectallocation id	Not null

13. SUBJECTALLOCATION TABLE

FIELD	TYPE	DESCRIPTION	CONSTRAINS
ID	Int	Subject allocation id	Primary key
SUBJECT_id	Int	Subject id	Not null, Foreignkey
semester	Varchar	Semester	Not null
COURSE_id	Int	Course id	Not null, Foreignkey

14. CLASSALLOCATION TABLE

FIELD	TYPE	DESCRIPTION	CONSTRAINS
ID	Int	Class allocation id	Primary key
COURSE_id	Int	Course id	Not null, Foreignkey
batch	Varchar	Batch	Not null
semester	Varchar	Semester	Not null
STAFF_id	Int	Staff id	Not null, Foreignkey

15. MATERIALS TABLE

FIELD	TYPE	DESCRIPTION	CONSTRAINS
ID	int	Material id	Primary key
name	varchar	Material name	Not null
date	varchar	Material date	Not null
time	varchar	Material time	Not null
SUBJECT_id	int	Subject id	Not null, Foreignkey

16. MANAGEMEETING TABLE

FIELD	TYPE	DESCRIPTION	CONSTRAINS
ID	int	Managemeeting id	Primary key
date	varchar	Meeting date	Not null
time	varchar	Meeting time	Not null
meeting	varchar	Meeting	Not null
batch	varchar	Batch	Not null
COURSE_id	int	Course id	Not null, Foreignkey
HOD_id	int	Hod id	Not null, Foreignkey

17. ATTENDANCE TABLE

FIELD	TYPE	DESCRIPTION	CONSTRAINS
ID	int	Attendance id	Primary key
time	varchar	Attendance time	Not null
date	varchar	Attendance date	Not null
status	varchar	Attendance status	Not null
STUDENT_id	int	Student id	Not null, Foreignkey
STAFFALLOCATION _id	int	Staffallocation id	Not null, Foreignkey

18. ASSIGNMENTNOTIFICATION TABLE

FIELD	TYPE	DESCRIPTION	CONSTRAINS
ID	int	Assignment id	Primary key
date	varchar	Assignment notification date	Not null
time	varchar	Assignment time	Not null
from_date	varchar	Starting date	Not null
to_date	varchar	Last date	Not null
topic	varchar	Assignment topic	Not null
SUBJECTALLOCATION _id	int	Subjectallocation id	Not null, Foreignkey

19. ASSIGNMENT TABLE

FIELD	TYPE	DESCRIPTION	CONSTRAINS
ID	int	Assignment id	Primary key
STUDENT_id	int	Student id	Notnull, Foreignkey
ASSIGNMENTNOTIFICATION_id	int	Assignment notification id	Notnull, Foreignkey
submission_date	varchar	Submission date	Not null
assignment	varchar	Assignment	Not null
status	varchar	Assignment status	Not null

20. QUESTIONPAPER TABLE

FIELD	TYPE	DESCRIPTION	CONSTRAINS
ID	int	Question paper id	Primary key
STAFF_id	int	Staff id	Notnull, Foreignkey
year	varchar	Question paper year	Not null
COURSE_id	int	Course id	Notnull, Foreignkey
qustioncode	varchar	Question paper code	Not null
qustions	Varchar	Question paper	Not null

7. SYSTEM TESTING AND IMPLEMENTATION

7.1 TESTING

System testing is an expensive but critical process that can take as much as 50% of the budget for program development. The common view of testing held by users is that it is performed to prove there are no errors in a program. However as indicated earlier, this is virtually impossible, since analyst cannot prove that software is free and clear of errors.

Therefore, the most useful and practical approach is with the understanding that the testing is the process of executing a program with explicit intention of finding errors that is making the program fail. The tester, who may be analysts, programmer or a specialist trained in software testing, is actually trying to make the program fail. In a successful test, then, one that find an error, analysts know that an effective testing program does not guarantee systems reliability. Therefore reliability must be designed in to the system.

7.2 TESTING STRATEGIES

As stated above the philosophy behind is to find errors. Test cases are devised with this purpose in mind. A test case is a set of data that system will process as normal input. However, the data are created with the express intend of determining whether the system will process them correctly. There are two general strategies for testing the software:

- Code testing
- Specification testing

CODE TESTING

The code testing strategy examines the logic of the program. To follow the testing method, we have to test cases that result in executing every instruction in the program or a module that is handled by the program On the surface code testing sems to be an ideal method for testing the software. But the rational that all the software errors can be unrecovered by checking every path in a program is faulty.

SPECIFICATION TESTING

To perform specification testing, the analysts examine the specifications stating what the program should do and how it should perform under various conditions. Then test cases are developed for each condition or combination of conditions and submitted for processing. By examining the results, the analyst can determine whether the program were a black box: the analyst does not look in to the program study the code and is not concerned about weather every instructions or path through the program is tested. In that case specification testing is not complete testing However, the assumption is that, if the program meets the specification, it will not fail.

7.3 PURPOSE OF TESTING

Testing verifies that software deliverable conforms precisely to the functional and design specifications that have been agreed to with the users. However, testing is also used in the detection of existing bugs. As a general rule, bugs are found and eliminated exponentially, that is, it gets harder to track down bugs as time goes by, but that doesn't mean that they aren't there. All tests should be traceable to believe that a logical path is not likely to be executed when it may be executed on regular basis. When a program is translated into programming language source code, it is likely that some typing errors will occur. Many will be uncovered by syntax and type checking mechanism but others may go undetected until testing begins.

Integration testing

Integration testing is a systematic for constructing the program structure while at the same time conducting tests to uncover errors associated with interfacing Unit tested module were taken and a single program structure was built that has been dictated by the design. Incremental integration has adopted here. The entire software was developed and tested in small segments, where errors were easy to locate and rectify. Program build (group of modules) were constructed corresponding to the successful testing of user interaction, data manipulation analysis, and display processing and database management. These tests can be performed in two ways:

- Top down integration

- Bottom up integration

Validation Testing

Validation testing is done to ensure complete assembly of the error-free software. Validation can be termed successfully only if it functions in manner that is reasonably expected by the customer. Under validation is alpha and beta testing. Alpha testing is where the end user tests the system rather than the developer. But in a controlled environment. The software is used on a natural setting with the developer monitoring the user using the system. The developer records the error and usage problems encountered by the user.

Program Testing

In this phase each program module is checked for two types of errors, syntax and logical. A syntax error is a program statement that violates one or more rules of the language. A logical error deals with in correct data fields, out of range items and invalid combinations. Thus each of every program statement is tested to check whether the expected output is attained.

String Testing

In this session, each program is tested to see whether it conforms to related programs in this system. Each portion is tested against the entire module with both test and live data before the entire system is ready to be tested.

System testing is conducted to uncover the weaknesses that were not found in earlier tests. This includes forced system failure and validation of the total system. In the earlier stage of system testing low volumes of input based on live data are entered. The volume is increased until the maximum level for each input type is reached. The total system is also tested for recovery and fallback

Integration testing is the next level up. It is concerned with conforming that no problems arise out of combining unit components in to more complex processes.

Window Design Testing

This kind of test proves that each individual window (such as primary, pop-up, dialog box, message box) that the system consists of secondary has been designed and built according to the project standards. The best method of ensuring this is in the form of a checklist- that is, points to be checked and signed off by the test viewer.

Black Box Testing

Although tests are designed to uncover errors, they are also used to demonstrate that the software functions are operational, input is properly accepted and output is correctly produced and that the integrity of external information id maintained. A black box test examines some of fundamental aspects of a system with little regard for the internal logical structure of the software.

All input screens were thoroughly tested for data validity and smoothness of data entry operations. Data entry operations are to be user friendly and smooth. It would be easier for the operators if they can enter data through keyboard only. Using mouse and keyboard alternatively will affect the speed of data entry. Care was taken to make data entry as smooth as possible. Also wrong inputs were tested for data values. For e.g.: entering alphabets for age, giving a non existing account number, giving incorrect username and password, entering numeric values that exceeds the upper or lower limits, wrong date format etc. all forms were tested for sample data.

White Box Testing

It is a test case design method that uses the control structure of the procedural design to derive test cases. Using the white box testing methods it was guaranteed that most of the independent paths within modules had been exercised at least once, all logical decisions on their true and false side executed all loops at their boundaries and exercised internal data structures to ensure their data validity.

White box testing has been done to achieve the following objectives. Logic errors and incorrect assumptions are inversely proportional to the probability that a program path will be executed. Errors tend to creep into the work when design and implementation function, condition or control that is out of the main stream. We often customer requirements. Testing should be planned before testing begins. Testing should begin "in the small" and progress towards testing "in the large".

Units/Component Testing

Unit testing focuses verification effort on smallest unit of software design module. Here, there are two modules namely, Admin, Registered users. In unit testing

- Module interface is tested to ensure that information properly lows into and put of the program under test.

- Local data structures are examined to ensure that data stored temporarily maintains its integrity during all steps in algorithm execution.
- Boundary condition is tested to ensure that the module operates properly at boundaries established to limit or restrict processing.
- All independent paths through the control structures are executed to ensure that all statements in the module have been executed at least once.
- Error handling paths are also tested.

Unit testing focuses verification efforts on the smallest unit of software design, the module. Here, the module interfaces, local data structures, boundary collections, and all the independent paths and last but not the least, all error handling paths were verified by inputting false data. Tests of data flow across each module interface of this software were done before any other test was initiated.

Navigation Testing

This test determines whether each window can be navigated to by mitiation (via all the multiple ways of testing) of all the different functions from any other appropriate window in the system, without necessarily performing any of the detailed processing that might be required to get the new system there.

7.4 DEBUGGING

Debugging occurs as a consequence of successful testing i.e. when a test case uncovers an error, debugging is the process that results in identifying the location of error and the removal of error. The poorly understood mental process that connects a symptom to cause is debugging. This process will always have one of the two outcomes:

- The cause will be found, corrected and then removed or
- The cause will not be found. In the latter case the person performing debugging may suspect a cause, design a test case to help validate his suspicion, and then work towards the correction of errors in the interactive fashion.

Following three approaches of debugging were used:

- Debugging by Indication
- Debugging by Deduction
- Backtracking

7.5 SYSTEM IMPLEMENTATION

Implementation is the process of converting newly received system into an operational one. It is the key stage in achieving a successful new system because usually it involves a lot of upheaval in the user department. It must therefore be carefully planned and controlled. Part from the planning the major two tasks of preparing implementation is education and training of users and testing of the system. Education of users of the system should take place much easier in the project that is when they are in the investigation and design work. Training has to be given to the staff regarding to the new system. Once the system has been trained the system can be test

Implementation is the stage in the project where the theoretical design is turned into a working system and is giving confidence on the new system for the users that it will work efficiently and effectively. It involves careful planning, investigation of the current system and it constraints on implementation, design of methods to achieve the change over, and evaluation, change over methods.

System implementation is putting the utility in to action. System implementations are the state in the project where theoretical design turned in to working system. The most crucial stage is to achieving a new successful system and giving confidence in new system that it will work efficiently and effectively. System implementation involves careful planning investigation of the current system and constraints on the implementation, design of methods to achieve.

Implementation procedures:

7.6Test plans: during test plans the elements to be tested in a planned, structured manner.

7.7 Training: the purpose of training is to ensure that all the personal who are to be associated with the computer based business system

7.8 Programmer training: programmers are assigned to the computer based business system. Project at the beginning of the development phase

7.9 Conversion: it is the process of performing all of the operations that result directly in the turnover of the new system to the user.

8. SECURITY TECHNOLOGIES AND POLICIES

The security features are considered while developing the system, so as to avoid the errors and omission that may lead to serious problems The system may have to face the unwanted events called threats Computer systems is secure against a particular threat if counter measures have been taken to reduce an acceptability low level amount of loss that the treat may be expected to cause over given period of time. A computer should be protected from the following three problems

- Loss of availability
- Loss of integrity
- Loss of confidentiality

These problems may adversely affect the ability of a computer system to carry out its intended task Threat to a computer system is any event that adversely affects the one or more assets or resources that make up the system. An event can be any of the following.

1. Interruption of communication
2. Destruction of hardware
3. Modification of software
4. Removal of programs
5. Disclosure of information

There are many methods for handling a threat

1. Avoid it by altering the design
2. Threat retention
3. Threat reduction i.e., the frequency of occurrence of a threat is reduced

There are many possible threats to the security and integrity of any system where more than one user is associated with the system. Software integrity has become increasingly important. The attribute measures a system's ability to withstand attacks, both accidental and intentional on its security Attacks can be made on all the three components of software: programs data and documents.

In this project the data security, data validation checking methods are applied using a password authentication. All the data, which is entered by the staff and administrator, will be validated. Users should enter their user name and password for the requesting access.

The system is also secured in such a way that even the slightest deviation in input the data will invoke error messages and provide necessary guidelines regarding the input. The result attained where fully satisfactory forms the customer point of view.

9. MAINTENANCE

The maintenance is important phase of any system. Maintenance of the system should be done accurately and with specific care proper running of the system.

Maintenance involves the software industry captive, typing up the system resources. It means restoring something to its original conditions Maintenance involves a wide range of activities including correcting, coding and design errors, updating documentation and test data and upgrading user support Maintenance is continued till the product is reengineered or deployed to another platform. Maintenance is also done based on fixing the problems reported, changing the interface with other software or hardware enhancing the software.

Any system developed should be secured and protected against possible hazards. The system should be maintained and upgraded according to the technological advancements. It ensures the data integrity, data control and security The system must be protected from fire and other natural calamities. The backup copies of data must be maintained daily so that to prevent the loss of data due to various reasons Security measures are provided to prevent unauthorized access of the database at various levels.

Software maintenance is divided into the following three categories

- Corrective maintenance
- Adaptive maintenance
- Perfective maintenance

Corrective maintenance has to do with the removal of residual errors present in the product when it is delivered as well as errors introduced into the software during its maintenance.

Adaptive maintenance modifies the software to keep it up to date with its operative environment .It may be needed because of changes in the user requirements, changes in target platforms, or changes in external interfaces.

Perfective maintenance involves changing the software to improve some of its qualities. The request to perfective maintenance may come directly from the software engineer, in order to improve the status of the product on the market, or they may come.

10. CONCLUSION

In conclusion, the upgraded college website for IHRD Neruvambram, Kannur, with its integrated AI chatbot, represents a significant step toward a more accessible and efficient information-sharing platform. By addressing the limitations of traditional systems and enhancing the user experience with real-time, automated support, the college is embracing a modern approach to communication. This digital transformation not only simplifies access to important information for students and faculty but also reflects the college's commitment to innovation and connectivity. Ultimately, this system empowers users, reduces administrative workload, and fosters an interactive online environment that aligns with the evolving needs of the college community.

11. SCOPE FOR FUTURE ENHANCEMENT

While the proposed digital platform offers significant improvements in accessing college-related information and support, there are several future enhancements that could further enrich the user experience and expand the platform's capabilities.

1. Integration with Learning Management Systems (LMS): By integrating with popular Learning Management Systems (such as Moodle, Blackboard, or Google Classroom), the platform could provide real-time updates on course materials, assignments, exam schedules, and grades. This would centralize both academic and administrative information in one accessible place for students and faculty.
2. Mobile App Development: Developing a mobile app version of the platform would ensure that users can access the information on the go. With the growing use of smartphones, a dedicated mobile app would provide an even more seamless experience, allowing users to get instant updates, notifications, and support anytime, anywhere.
3. Personalized Dashboard: Future updates could include personalized dashboards for students, faculty, and visitors, which display customized content based on their roles and preferences. For example, students could have access to their courses, schedules, grades, and upcoming events, while faculty could have access to course

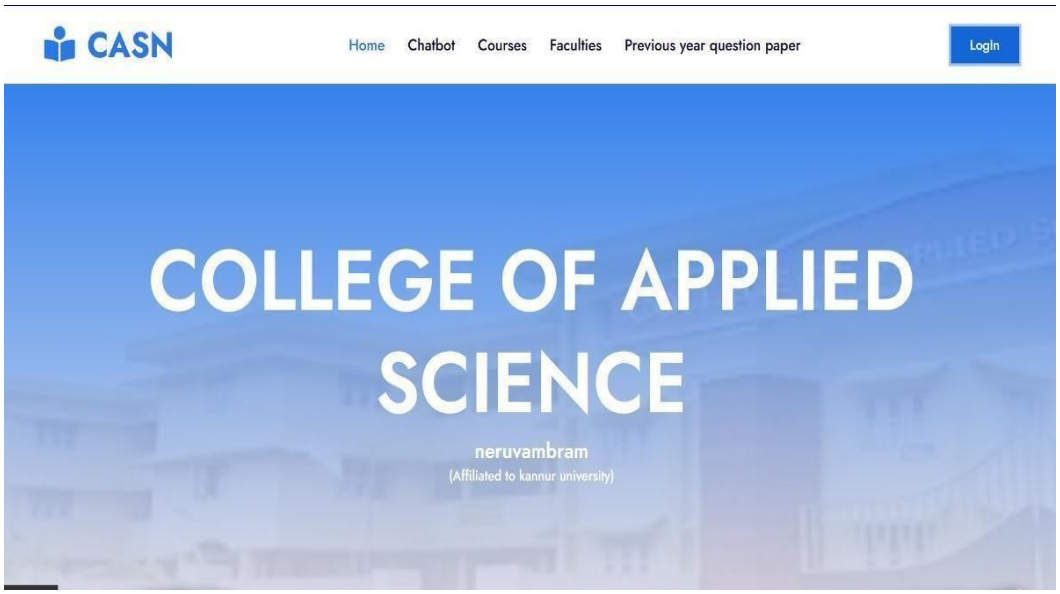
management tools, student performance data, and other faculty-specific resources.

4. Advanced AI Features: The AI chatbot could be enhanced with natural language processing (NLP) to better understand and respond to a wider variety of user queries. It could also be trained to handle more complex requests, such as assisting with course selection, career guidance, or even offering mental health support and counseling resources.
5. Virtual Assistant for Administrative Tasks: The platform could include a virtual assistant to help with administrative tasks such as applying for leave, checking fee payments, or requesting transcripts. This would save time for both students and administrative staff.
6. Integration with Social Media and Campus Events: To improve engagement, the platform could integrate with social media channels and provide updates about campus events, news, or achievements. This would keep the entire campus community informed and connected, fostering a sense of belonging.
7. Online Collaboration Tools: The addition of features such as real-time document sharing, discussion forums, and collaborative project spaces could enable students and faculty to work together on academic projects more efficiently, even remotely.
8. Analytics for Institutional Improvement: With the collection of user interaction data, the platform could generate analytics to help the college improve its services. For example, identifying commonly asked questions or user behavior patterns could help the institution improve its offerings, optimize resources, and enhance the overall experience for the community.

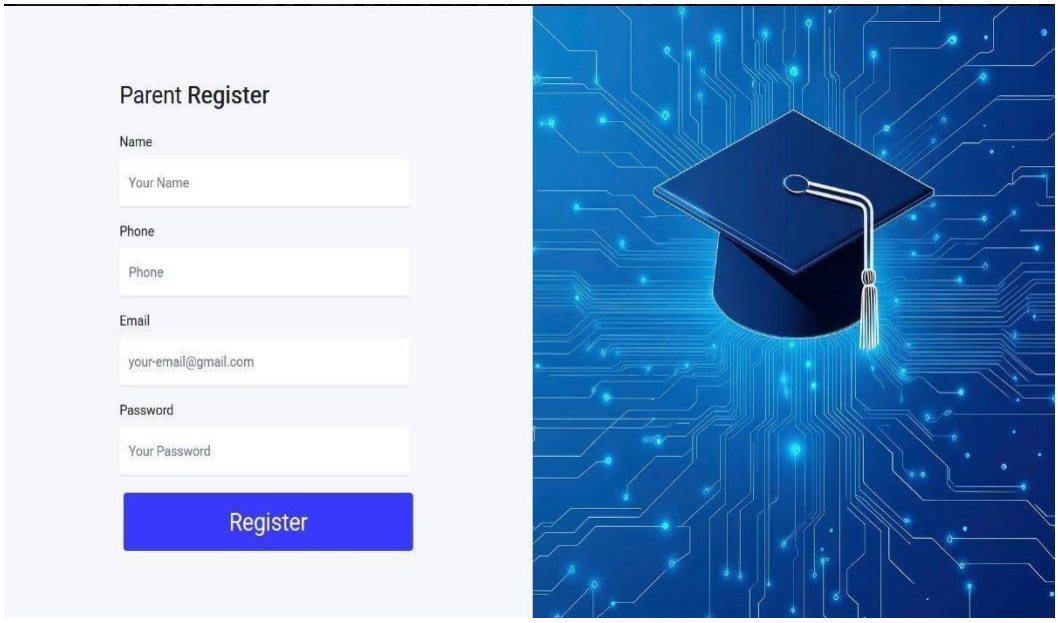
By incorporating these enhancements, the digital platform could evolve into an even more powerful tool that caters to the evolving needs of IHRD Neruvambram, fostering a more connected, efficient, and innovative academic environment.

12. SCREENSHORT

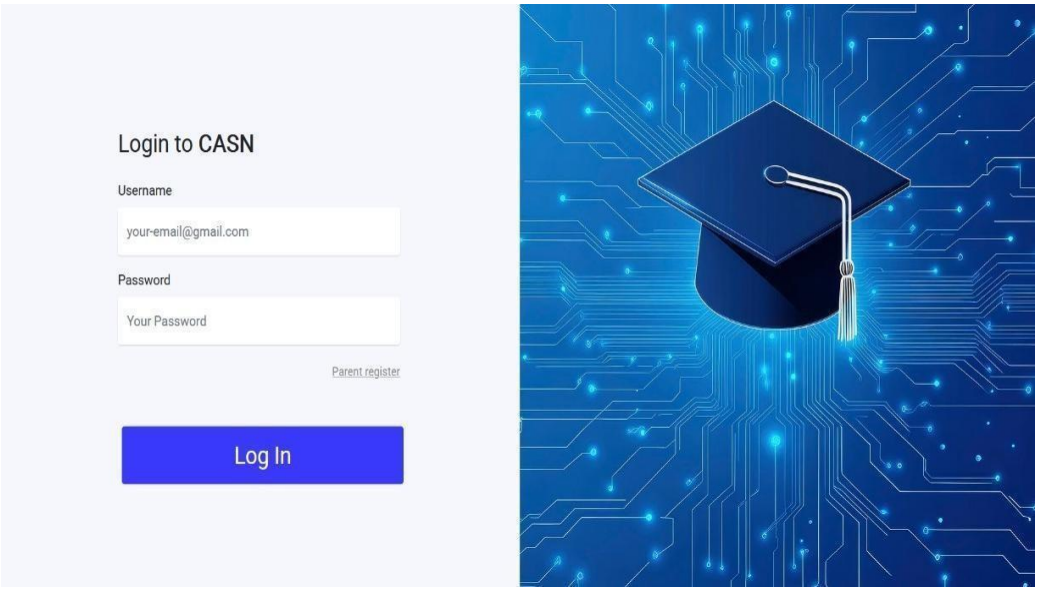
- **Default home page**



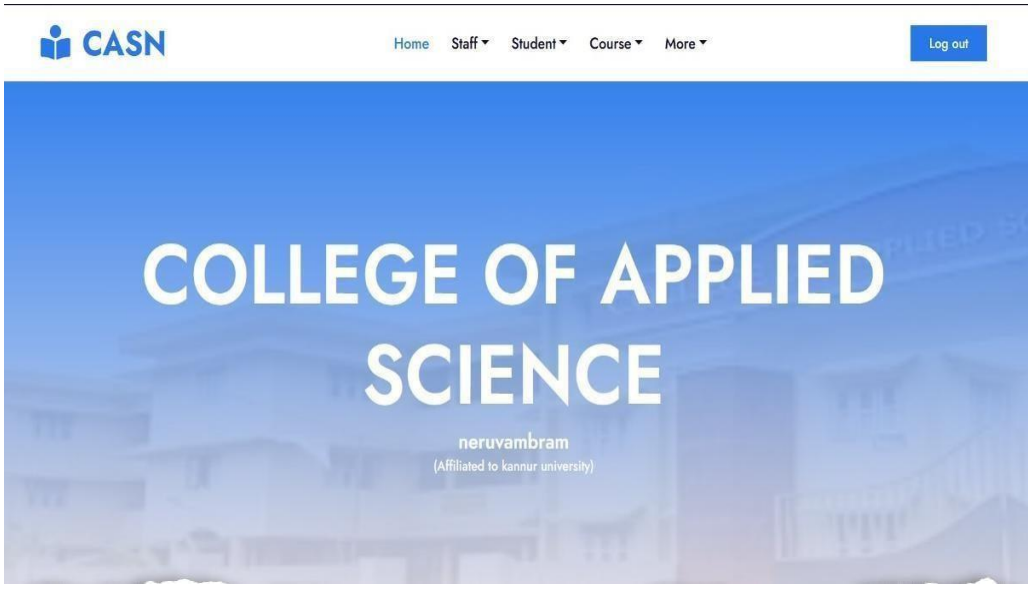
- **Register page for Parent**



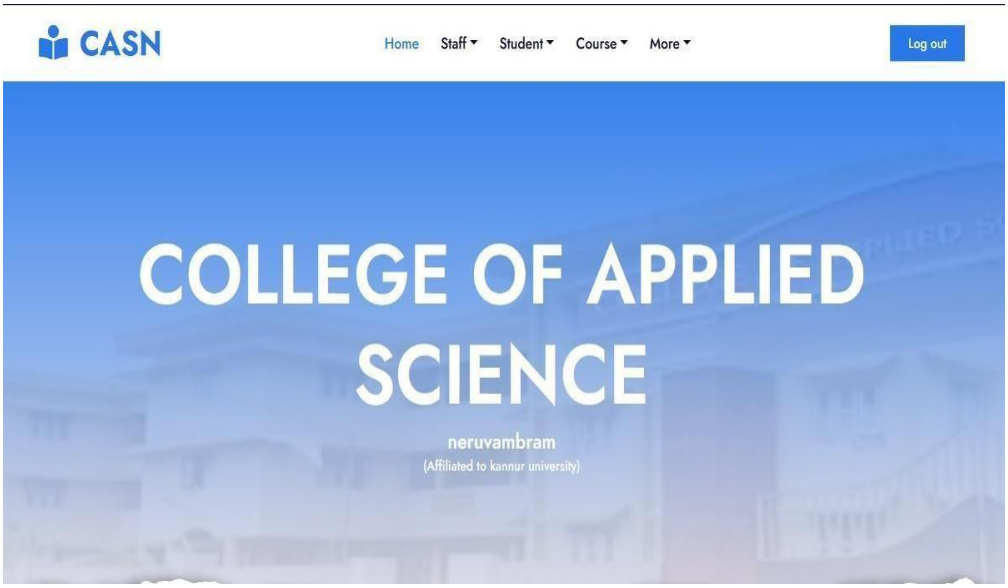
- **Login page for user**



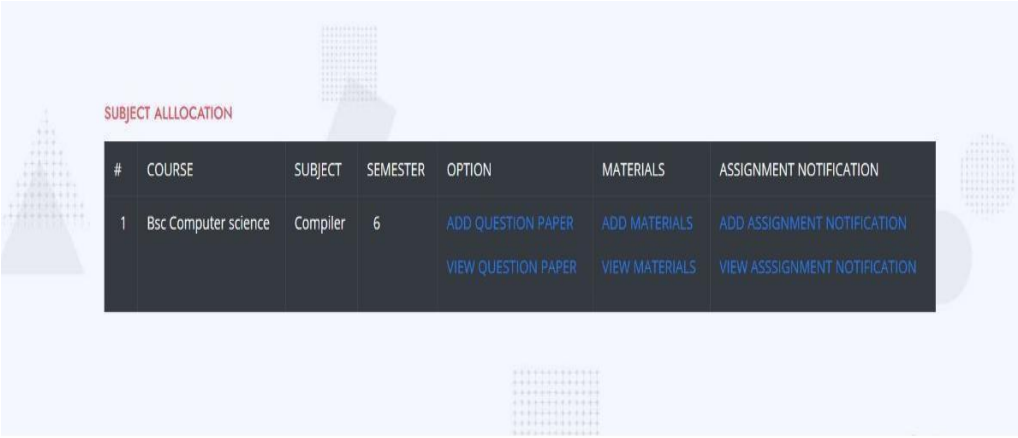
- **Admin home page**



- **Staff details page for admin**



- **Subject allocation page for staff**



- **Assignment notification page for staff**

ASSIGNMNET NOTIFICATION

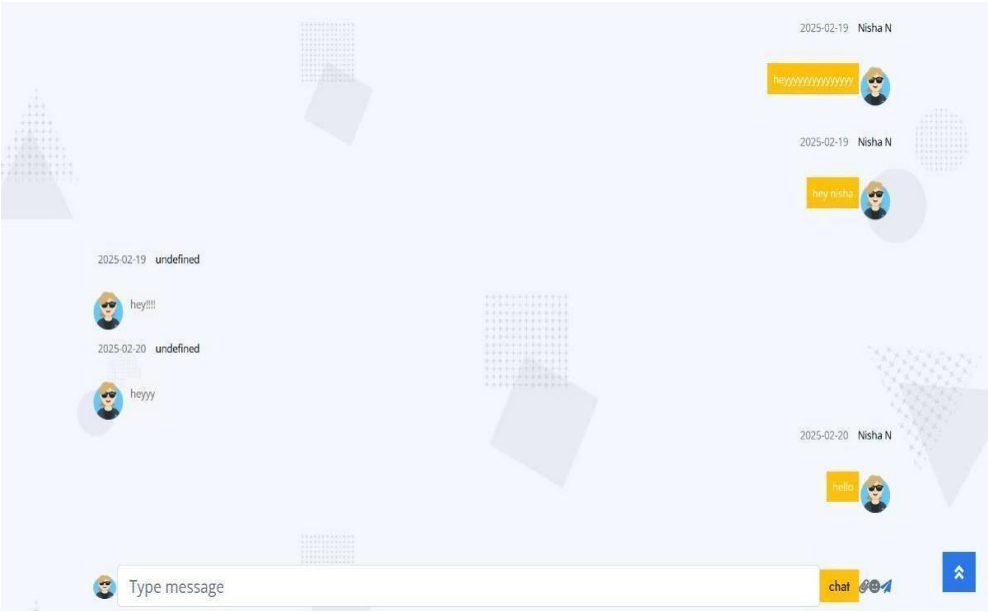
#	DATE	TIME	FROM_DATE	TO_DATE	TOPIC	OPTION
1	2025-02-12	12:33	2025-02-12	2025-02-14	symbol tableee	EDIT DELETE
2	2025-02-13	10:30	2025-02-20	2025-03-06	symbol table	EDIT DELETE

- **Verified parent page for staff**

VERIFIED PARENT

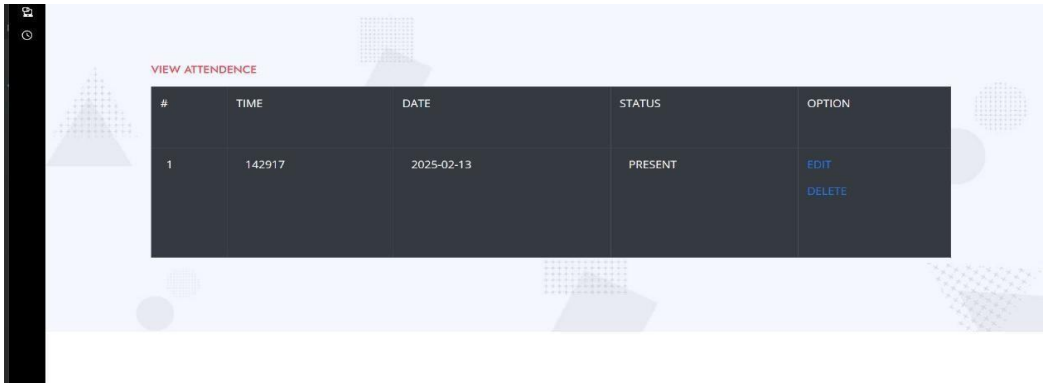
#	STUDENT NAME	COURSESNAME	BATCH	PARENT NAME	PHONE	OPTION
1	Theertha N	Bsc Computer science	2022-25	Nisha N	9995908992	Chat
2	Chinmaya v v	Bsc Computer science	2022-25	Savithri V V	9544951799	Chat

- **Chat between parent and staff**

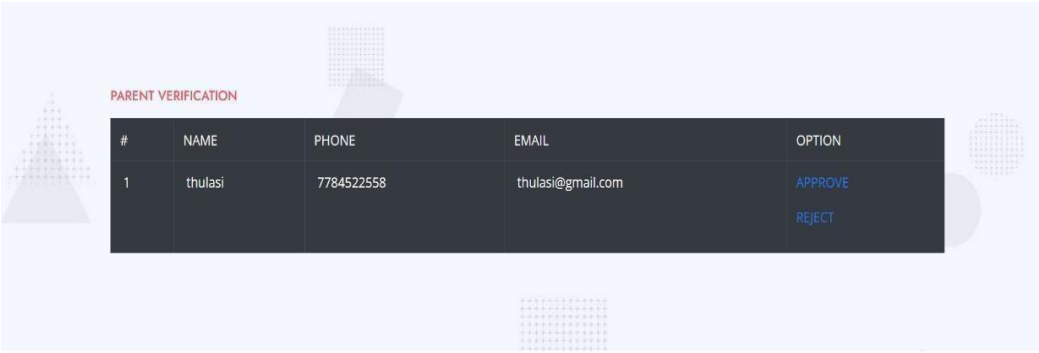


- **Class in charge allocation page for hod**

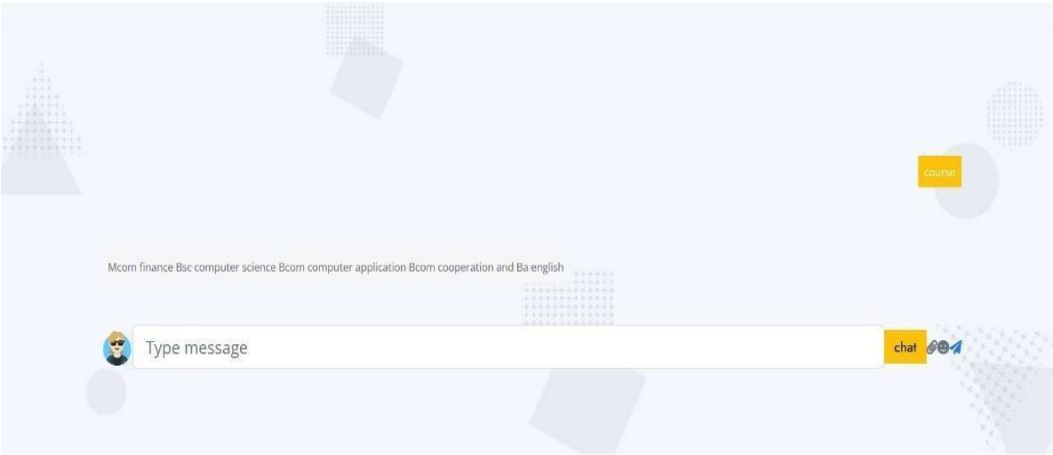
- **View attendance page for parent**



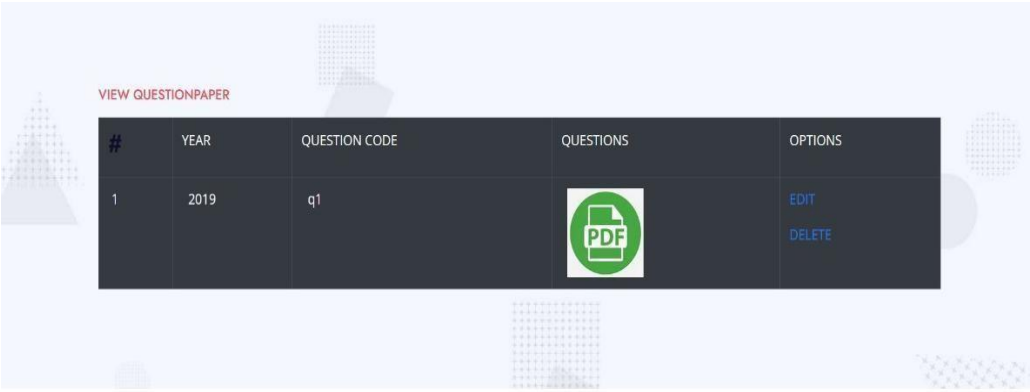
- **Parent verification page for staff**



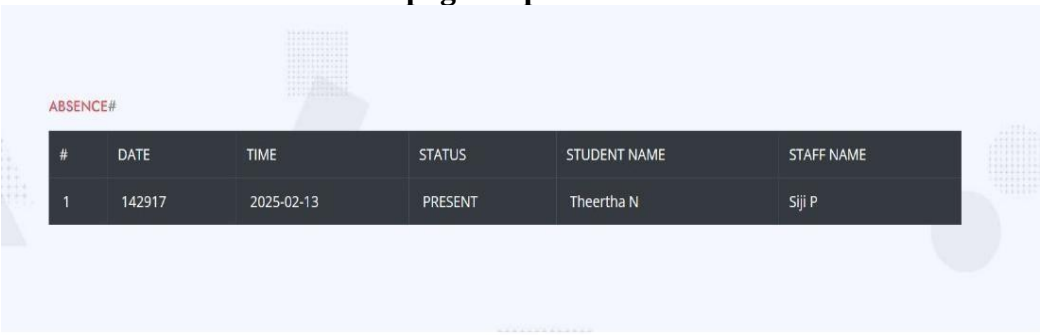
- **Chatbot for public**



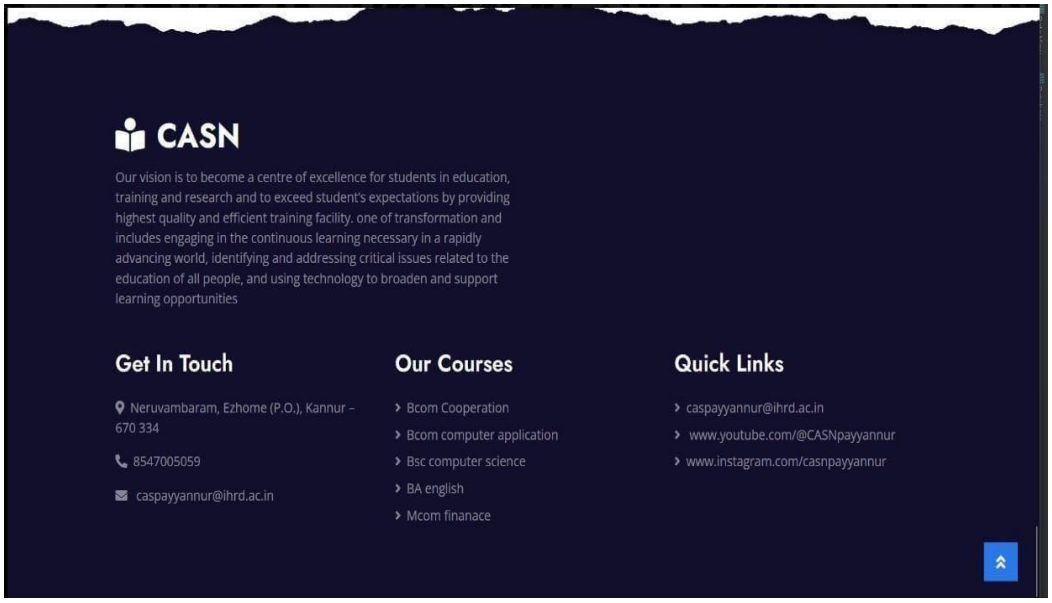
- **View previous question paper for public**



- **Attendance page for parent**

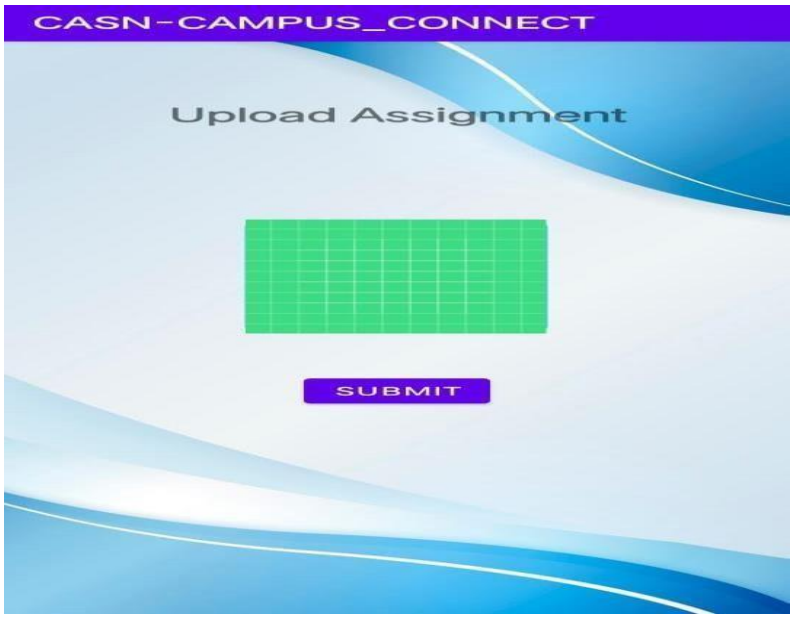


- **Contact page for public**



- **Home page for student(Android)**

- **Upload assignment page for student**



13.REFERENCES AND BIBILOGRAPHY

- https://chat.openai.com
- https://google.com
- https://www.youtube.com